VLADIMIR DVORKIN

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CONTRIBUTIONS

Working papers

V. Dvorkin, F. Fioretto, P. Van Hentenryck, J. Kazempour, P. Pinson, "**Differentially Private Convex Optimization with Feasibility Guarantees**,"
Submitted to the 34th Conference on Neural Information Processing Systems (NeurIPS)

V. Dvorkin, A. Ratha, J. Kazempour, P. Pinson, "**Stochastic Control and Market Design for Natural Gas Networks**,"
Available upon request

V. Dvorkin, J. Kazempour, P. Pinson, "**Spiteful Bidding in Electricity Markets**,"
Available upon request

Published papers

V. Dvorkin, F. Fioretto, P. Van Hentenryck, J. Kazempour, P. Pinson, "**Differentially Private Optimal Power Flow for Distribution Grids**," IEEE Transactions on Power Systems, 2020

V. Dvorkin, P. Van Hentenryck, J. Kazempour, P. Pinson, "**Differentially Private Distributed Optimal Power Flow**," In: 59th Conference on Decision and Control, 2020

V. Dvorkin, J. Kazempour, P. Pinson, "**Chance-Constrained Equilibrium in Electricity Markets With Asymmetric Forecasts**," In: 16th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), 2020

V. Dvorkin, S. Delikaraoglou, J.M. Morales, "**Setting Reserve Requirements to Approximate the Efficiency of the Stochastic Dispatch**," IEEE Transactions on Power Systems, 2019.

V. Dvorkin, J. Kazempour, P. Pinson. "**Electricity Market Equilibrium under Information Asymmetry**." Operations Research Letters, 2019

A.M. Radoszynski, V. Dvorkin, P. Pinson, "**Accommodating Bounded Rationality in Pricing Demand Response**," In: 2019 IEEE Milan PowerTech. IEEE, 2019

V. Dvorkin, J. Kazempour, L. Baringo, P. Pinson, "**A Consensus-ADMM Approach for Strategic Generation Investment in Electricity Markets**," In: IEEE 57th Annual Conference on Decision and Control (CDC), 2018

WORK EXPERIENCE

Visiting Scholar
Georgia Institute of Technology Jul 2019 – Dec 2019 Atlanta, USA

- School of Industrial and Systems Engineering (ISyE)
- Supervised by Pascal Van Hentenryck
- Development of differentially private optimization algorithms

Research Assistant
Higher School of Economics Dec 2013 - Aug 2017 Moscow, Russia

- Institute of Pricing & Regulation of Natural Monopolies
- Supervised by Dr. Iliya Dolmatov
- Electricity, gas & heat markets
- Projects with Energy Ministry of Russia, Federal Tariff Service, Transmission and Distribution Companies

Consultant
Khaznah Strategies Ltd. May 2017 - Aug 2017 London, UK

- Energy and natural resources price forecasting
- Software Engineering

Engineering Intern
Power Engineering Group EOL Sep 2011 - Nov 2013 Moscow, Russia

- Designing high-voltage circuits for consumer electronics
- Developing the high-voltage devices for mass production.

TEACHING EXPERIENCE

Renewables in Electricity Markets
M.Sc. Spring 2020 Technical University of Denmark

M.Sc. Spring 2017 Technical University of Denmark

Advanced Optimization in Electricity Markets
M.Sc./Ph.D Fall 2018 Technical University of Denmark

Decomposition techniques for energy systems applications
M.Sc./Ph.D Fall 2018 Skolkovo Institute of Science and Technology

EDUCATION

Ph.D. in Electrical Engineering
Technical University of Denmark (DTU)
Sep 2017– Nov 2020 Denmark, Lyngby

- Project title: Advanced Game-Theoretical Aspects in Electricity Markets
- Supervised by Prof. Pierre Pinson and Prof. Jalal Kazempour

M.Sc. in Sustainable Energy
Technical University of Denmark (DTU)
Aug 2015 – Jun 2017 Denmark, Lyngby

- Primaries: power systems stability & control, optimization, decision-making, game theory, energy economics&policy

M.Sc. in Energy Economics
Higher School of Economics (HSE)
Sep 2012 – Jun 2014 Russia, Moscow

- Primaries: microeconomics, finance, fundamentals of energy economics, corporate management

B.E. in Electrical Engineering
Moscow Power Engineering Institute (MPEI)
Sep 2008 – Jun 2012 Russia, Moscow

- Primaries: power systems control, planning & optimization

LANGUAGES

Russian
English

SKILLS

Julia | Python | MATLAB | GAMS | Optimization | Equilibrium analysis | Decomposition | ANN | Regression analysis | LaTeX | Forecasting | Clustering | Data Privacy

SCHOLARSHIPS & AWARDS

LANL Grid Science Winter School Scholarship
Jan 2019

Outstanding Reviewer, IEEE Trans. Sustain. Energy
2018

DTU Tuition Fee Waiver for MSc Students
Aug 2015 - Jun 2017

HSE Scholarship for Science Achievements
March 2014

HSE Scholarship for Excellency
Sep 2012- Jun 2014

Semifinalist (10/300) at the Youth Russian Petroleum&Gas case championship
Jun 2013

MPEI Scholarship for Academic Achievements
Sep 2008 - Dec 2012

PRESENTATIONS

Stochastic Control and Market Design for Natural Gas Networks
- Massachusetts Institute of Technology — Sep 2020

Differentially Private Optimization of Power Systems
- 2020 INFORMS Annual Meeting — Oct 2020
- 2019 Energy Systems and Optimization Workshop — Nov 2019
- ISyE Student Seminars — Dec 2019

Electricity Market Equilibrium under Information Asymmetry
- 2019 INFORMS Annual Meeting — Oct 2019
- 2019 IEEE PES General Meeting — Aug 2019
- XV International Conference on Stochastic Programming (ICSP2019) — Aug 2019
- Grid Science Winter School, Los Alamos National Laboratory — Jan 2019
- Johns Hopkins University — Jan 2019

A Consensus-ADMM Approach for Strategic Generation Investment in Electricity Markets
- 2018 IEEE Conference on Decision and Control (CDC) — Dec 2018
- Computational Management Science (CMS) Conference 2018 — May 2018

REVIEWER EXPERIENCE

- European Journal of Operational Research — Since Jan 2020
- IEEE Transactions on Smart Grids — Since Apr 2019
- IEEE Transactions on Automatic Control — Since Jan 2019
- IEEE Transactions on Sustainable Energy — Since Jun 2018
- IEEE Transactions on Power Systems — Since Mar 2018
- International Transactions on Electrical Energy Systems — Since Oct 2017
- IEEE Power Systems Computation Conference — 2018,2020
- IEEE Conference on Decision and Control — 2018-2020
- Smart Energy Systems and Technologies (SEST) — 2020
- IEEE PES PowerTech — 2019
- IEEE American Control Conference — 2018

REFERENCES

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